

1.1. E-Commerce architecture

E-Commerce Architecture is the structure of an e-commerce system that defines how different components interact to provide online services such as product browsing, ordering, payment, and inventory management. A common e-commerce system uses Multi-Tier Architecture, especially Three-Tier Architecture, where the application is divided into separate layers.

1. Presentation Layer

The Presentation Layer is the front-end of the e-commerce system where customers interact with the application through a website or mobile app. It displays product information, search results, shopping carts, login pages, checkout forms, and other user-interface elements. It is commonly developed using HTML, CSS, JavaScript, and frameworks such as React, Angular, or Vue. This layer sends user requests to the Business Logic Layer through HTTP/HTTPS and displays the returned results.

2. Business Logic Layer

The Business Logic Layer is the middle tier and acts as the brain of the e-commerce system. It receives requests from the Presentation Layer, processes them according to business rules, and communicates with the Data Layer when information needs to be stored or retrieved. It handles tasks such as user authentication, order processing, price calculation, discounts, payment processing, inventory checking, and shipping management. Technologies such as Django, Flask, Spring, Laravel, and Node.js can be used to implement this layer.

3. Data Layer

The Data Layer is responsible for the permanent storage and management of e-commerce data. It stores information such as customer accounts, products, orders, inventory, payments, and reviews. Relational databases such as MySQL, PostgreSQL, and Oracle are commonly used, while technologies such as MongoDB and Redis may be used for NoSQL data or caching. For security, users cannot directly access the database; the Business Logic Layer communicates with it on their behalf.

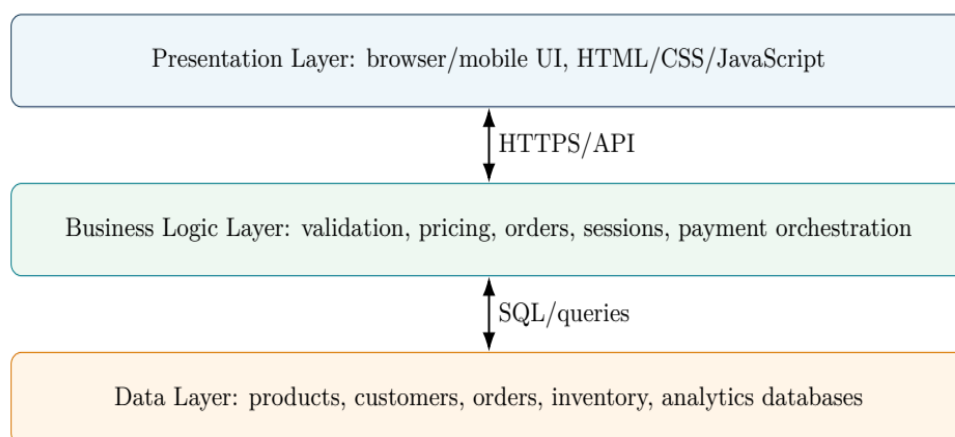
Architecture Flow

Customer → Presentation Layer → Business Logic Layer → Data Layer

The response follows the reverse direction:

Data Layer → Business Logic Layer → Presentation Layer → Customer

For example, when a customer searches for a product, the Presentation Layer sends the request to the Business Logic Layer. The Business Logic Layer processes the request and retrieves the required product information from the Data Layer. The result is then returned to the Presentation Layer and displayed to the customer.



1.2 Why Separate an E-Commerce System into Tiers? (Optional)

Separating an e-commerce application into different tiers provides several important benefits:

1. Security

- The database is not directly accessible from the internet.
- Users communicate with the application server, which then accesses the database.
- This provides an additional layer of protection.

2. Scalability

- Each tier can be scaled independently.
- For example, during a large sale, more application servers can be added to handle increased traffic without changing the database or front-end.

3. Maintainability

- Changes in one tier usually do not require major changes in other tiers.
- For example, the website design can be updated without changing the business logic.

4. Parallel Development

- Different teams can work on different tiers simultaneously.
- Front-end developers → Presentation Layer
- Back-end developers → Business Logic Layer
- Database administrators → Data Layer

5. Reusability

- The same business logic can serve multiple clients.
- For example, one backend can support a website, mobile app, and external API.

1.3 E-Commerce Business Models

A business model is a plan that explains how an organization creates, delivers, and captures value. In e-commerce, it describes who the buyer and seller are, how they interact, and how the business earns money. The major e-commerce business models are B2C, B2B, C2C, C2B, B2G, and G2C. M-Commerce and Social Commerce are important channels through which these models can operate.

1. B2C – Business-to-Consumer

Business-to-Consumer (B2C) is an e-commerce business model in which a business sells products or services directly to individual consumers for their personal use through an online website, application or digital marketplace. **Example:** An online electronics company sells a smartphone directly to an individual customer through its website.

2. B2B – Business-to-Business

Business-to-Business (B2B) is an e-commerce business model in which one business sells products, services, raw materials, or other resources directly to another business through an online platform, usually for business operations, production, or resale. **Example:** A wholesale electronics company sells 100 laptops to a computer retailer through an online wholesale platform.

3. C2C – Consumer-to-Consumer

Consumer-to-Consumer (C2C) is an e-commerce business model in which one individual consumer sells products or services directly to another individual consumer through an online marketplace or platform that connects buyers and sellers. **Example:** An individual sells a used laptop to another individual through Facebook Marketplace.

4. C2B – Consumer-to-Business

Consumer-to-Business (C2B) is an e-commerce business model in which an individual consumer offers or sells products, services, skills, content, or other value to a business through an online platform, and the business pays the individual for it.

Example: An individual graphic designer creates and sells a logo design to a company through Fiverr.

5. B2G – Business-to-Government

Business-to-Government (B2G) is an e-commerce business model in which a private business sells products or provides services directly to a government agency, department, or public organization through electronic procurement, bidding, or online tender systems.

Example: A software company develops and provides a digital management system to a government department through an online tender.

6. G2C – Government-to-Citizen

Government-to-Citizen (G2C) is an electronic service model in which a government or government agency provides public information, facilities, or services directly to individual citizens through websites, applications, or other digital platforms.

Example: A citizen applies for a passport, pays the required fee, and checks the application status through a government online portal.

Other Important E-Commerce Channels

8. M-Commerce — Mobile Commerce

M-Commerce means conducting commercial transactions through mobile devices such as smartphones and tablets. It is not a separate business model. Rather, it is a channel through which B2C, B2B, C2C, and other models can operate.

Examples include mobile shopping apps, mobile payments, super-apps, and location-based commerce.

Example: Purchasing a product through a shopping app or paying a merchant through a mobile financial service.

9. Social Commerce

Social Commerce means buying and selling products or services through social media platforms.

It includes shoppable posts, live-stream selling, influencer-driven sales, community group buying, and social proof such as ratings and customer reviews.

Example: A seller demonstrates clothes through Facebook Live and customers place orders during the stream.

1.4. E-Commerce vs F-Commerce

E-Commerce

E-Commerce (Electronic Commerce) refers to buying and selling products or services through dedicated websites, mobile apps, or online marketplaces. Examples: Daraz, Amazon, an online store's own website.

Customers usually search for products, add them to a cart, make payments, and complete orders through a structured digital system.

F-Commerce

F-Commerce (Facebook Commerce) is buying and selling products or services through Facebook. It is a specific form of Social Commerce. Examples: Facebook Marketplace, Facebook business pages, Facebook groups, and Facebook Live selling.

In Bangladesh, sellers may promote products through Facebook and complete orders through Messenger, Cash on Delivery (COD), or Mobile Financial Services (MFS).

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Feature	E-Commerce	F-Commerce
Platform	Uses online shopping websites, apps, or marketplaces	Mainly uses Facebook pages, groups, and Marketplace

Product Discovery	Customers find products by searching, browsing categories, or seeing ads	Customers find products from Facebook posts, groups, ads, Live videos, or shared content
Trust	Customers trust the seller based on brand name, ratings, reviews, and buyer protection	Customers trust sellers based on Facebook profiles/pages, reviews, comments, and community feedback
Transaction	Customers add products to a cart, checkout, and pay	Customers often message the seller, place an order, and pay by COD or mobile payment
Interaction	Communication is usually formal and focused on buying and selling	Communication is more friendly, direct, and conversational
Setup	Usually needs a website, app, or seller account on a marketplace	Can easily start with a Facebook page, profile, or group

2. E-Commerce Revenue Models with example

An E-Commerce Revenue Model explains how an online business earns money from its products, services, platform, or users. A company can use more than one revenue model simultaneously. For example, a marketplace may earn from sales commissions, subscriptions, and advertising at the same time.

1. Sales

The Sales or E-Tail model is a revenue model in which a business earns money by selling physical products, digital products, or services directly to customers through an online website, app, or marketplace. Revenue is generated whenever a customer purchases a product or service.

Example: Amazon sells a laptop to a customer online and earns revenue from the amount paid for the laptop.

2. Subscription Model

The Subscription model is a revenue model in which customers pay a fixed amount regularly, such as monthly or yearly, to access a product, service, content, or special features for a specific period of time. The business continues earning revenue as long as customers keep their subscriptions.

Example: Netflix charges users a monthly fee to access movies, TV shows, and other content.

3. Transaction Fee Model

The Transaction Fee model is a revenue model in which a business earns money by charging a fee or commission for each transaction completed through its website, app, or online platform. The company usually acts as an intermediary between buyers and sellers.

Example: An online marketplace charges a seller a commission whenever a product is successfully sold through the platform.

4. Advertising Model

The Advertising Model is a revenue model where a business earns money by showing advertisements from other companies on its website, app, or online platform. Companies pay for these ads based on how many people see, click, or interact with them.

Example: YouTube shows advertisements before or during videos and earns money from the companies that place those advertisements.

5. Freemium Model

The Freemium model is a revenue model in which a business provides basic features or services to users for free while charging money for premium features, additional services, more storage, or an improved experience. The free version attracts users, while some users upgrade to the paid version.

Example: Spotify provides free music streaming with advertisements but charges users for Premium features such as ad-free listening and offline downloads.

6. Affiliate

The Affiliate or Referral model is a revenue model in which a person or business earns a commission by promoting another company's products or services and referring potential customers to that company, usually through special links or referral codes. The commission is earned when the referred customer completes a required action, such as making a purchase.

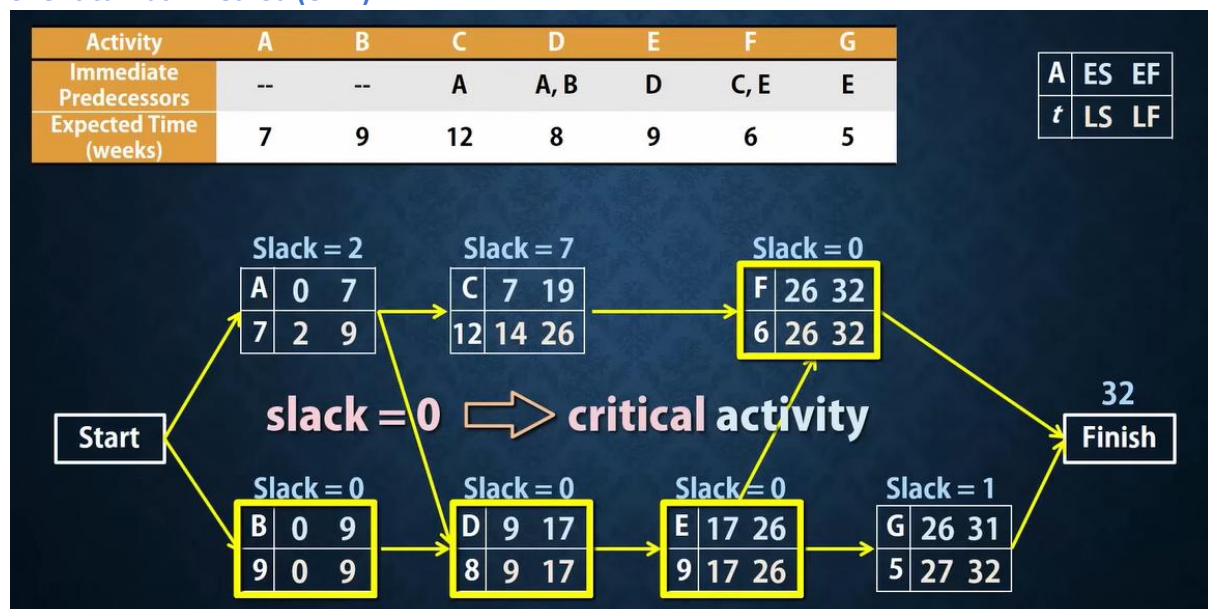
Example: A blogger shares an affiliate link to a laptop. If a reader clicks the link and buys the laptop, the blogger receives a commission from the seller.

Amazon: Multi-Revenue Model Example

Amazon is a useful example because its ecosystem combines several revenue sources. It earns through direct retail sales, fees from third-party marketplace sellers, Prime subscriptions, advertising, AWS cloud services, and its affiliate program.

This demonstrates an important point: a business does not need to depend on only one revenue model. Large e-commerce platforms often combine several models to diversify and increase revenue.

3. Critical Path Method (CPM)



Formula

Forward Pass → Take Maximum

The forward pass calculates **Earliest Start (ES)** and **Earliest Finish (EF)**.

If an activity has **multiple predecessors**, take the **maximum EF** of them:

$$ES = \max(\text{EF of predecessors})$$

Why? The activity cannot start until **all previous activities are finished**.

Example: Previous activities finish at 5, 8, and 6.

$$ES = \max(5, 8, 6) = 8$$

Backward Pass → Take Minimum

The backward pass calculates Latest Finish (LF) and Latest Start (LS).

If an activity has multiple successors, take the **minimum LS** of them:

$$LF = \min(\text{LS of successors})$$

Example: Next activities have LS values 12, 10, and 15.

$$LF = \min(12, 10, 15) = 10$$

Slack Calculation:

$$\boxed{Slack = LS - ES}$$

or

$$\boxed{Slack = LF - EF}$$

Critical path = B-D-E-F

4. Michael Porter's Value Chain Model

Michael Porter's Value Chain Model is a framework used to understand how a company creates value for its customers through different business activities. A company can gain a competitive advantage by performing these activities better, faster, or at lower cost than its competitors. Porter divides business activities into Primary Activities and Support Activities.

1. Primary Activities – Directly Create Value

Primary activities are the main activities directly involved in creating, selling, delivering, and supporting a product or service.

Inbound Logistics: Activities related to receiving, storing, and managing materials or products coming into the company. In e-commerce, this can include receiving products from suppliers, checking incoming stock, and updating inventory systems.

Operations: Activities that turn inputs into finished products or prepare products for customers. In e-commerce, this includes processing orders, managing warehouses, packaging products, and using automated systems or robots.

Outbound Logistics: Outbound Logistics refers to the activities involved in storing, packing, shipping, and delivering finished products to customers. Its main goal is to ensure that products reach customers quickly, safely, and on time.

Marketing & Sales: Activities that attract customers and encourage them to buy products or services. Examples include online advertising, SEO, social media marketing, email marketing, and product recommendations.

Service: Activities that help customers after they purchase a product or service and maintain customer satisfaction. Examples include customer support, online returns, chatbots, warranty services, and loyalty programs.

2. Support Activities – Help Primary Activities

Support activities provide the resources, technology, people, and management needed for primary activities to work effectively.

Firm Infrastructure: Activities related to the overall management and administration of the company, including finance, accounting, legal services, planning, and information systems. For example, an e-commerce company may use ERP systems and cloud services to manage its operations.

Human Resource Management (HRM): Activities related to recruiting, training, managing, and paying employees. E-commerce businesses may hire software engineers, data scientists, marketing specialists, and UX designers.

Technology Development: Activities related to developing and improving technology, software, products, and business processes. Examples include developing websites and apps, improving recommendation systems, and automating warehouse operations.

Procurement: Activities related to purchasing the products, materials, equipment, and services needed by the company. E-commerce businesses can use online procurement systems and supplier portals to make purchasing easier and faster.

Firm Value Chain vs. Industry Value Chain

The Firm Value Chain describes the activities inside one company that create value, from receiving materials to providing after-sales service. The Industry Value Chain describes the activities of all businesses involved in bringing a product from raw materials to the final customer, such as manufacturers, wholesalers, retailers, and delivery companies.

Disintermediation happens when e-commerce removes traditional middlemen such as wholesalers or distributors. For example, a manufacturer can sell directly to customers through its website.

Re-intermediation happens when new digital middlemen are introduced. For example, online marketplaces can connect sellers with customers and provide services such as payments, reviews, and delivery support.

MIS Integration in the Value Chain

Management Information Systems (MIS) can improve different activities in the value chain by providing accurate information, automation, and better decision-making.

For example, in Inbound Logistics, supplier and inventory systems can reduce stock shortages and purchasing costs. In Operations, MIS can improve production scheduling and warehouse management. In Marketing & Sales, CRM systems can help businesses understand customers and target them with suitable offers. In Service, helpdesk and customer-support systems can solve customer problems faster and improve customer satisfaction.

5. Subscription Model vs Freemium Model

The Subscription Model is a revenue model in which customers pay a regular fee, such as monthly or yearly, to continuously access a product, service, or content. Example: Netflix charges users a monthly fee to access movies and TV shows.

The Freemium Model is a revenue model in which a business provides basic features or services for free but charges users for premium features, additional services, or an improved experience. Example: Spotify provides basic music streaming for free, but users pay for Premium to get features such as ad-free listening and offline downloads.

Subscription vs Freemium

Feature	Subscription Model	Freemium Model
Basic Idea	Pay regularly to use the service	Basic service is free; advanced features are paid
Payment	Monthly or yearly fee	Payment only when users upgrade
Free Access	Usually not available, except trials	Basic version is available for free
Revenue Source	Regular subscription fees	Premium upgrades from free users
Main Goal	Generate stable and recurring revenue	Attract many free users and convert some to paid users
Example	Netflix	Spotify

6. Key E-Commerce Metrics and Formulas

E-commerce metrics are measurements used to understand how well an online business is performing. They help businesses measure sales, advertising performance, customer behavior, costs, inventory, and long-term growth.

Conversion Rate (CR)

Conversion Rate shows the percentage of website visitors who make a purchase.

$$CR = \frac{\text{Purchases}}{\text{Visits}} \times 100$$

Example: If 1,000 people visit a website and 30 buy something:

$$CR = \frac{30}{1000} \times 100 = 3\%$$

So, 3% of visitors became customers.

Cost Per Click (CPC)

CPC shows how much a business pays on average for each click on an advertisement.

$$CPC = \frac{\text{Ad Spend}}{\text{Number of Clicks}}$$

Example: If a company spends 5,000 BDT on ads and receives 1,000 clicks:

$$CPC = \frac{5000}{1000} = 5 \text{ BDT}$$

The business pays 5 BDT per click.

Customer Acquisition Cost (CAC)

CAC shows how much a business spends to gain one new paying customer.

$$CAC = \frac{\text{Total Marketing Spend}}{\text{New Customers Acquired}}$$

Example: If a business spends 50,000 BDT on marketing and gains 100 customers:

$$CAC = \frac{50,000}{100} = 500 \text{ BDT}$$

It costs 500 BDT to acquire each new customer.

Average Order Value (AOV)

AOV shows the **average amount of money customers spend on each order**.

$$AOV = \frac{\text{Total Revenue}}{\text{Number of Orders}}$$

Example: If a business earns 100,000 BDT from 200 orders:

$$AOV = \frac{100,000}{200} = 500 \text{ BDT}$$

The average customer spends 500 BDT per order.

Customer Lifetime Value (CLV)

CLV shows the total revenue a business expects to earn from one customer during the entire time that person remains a customer.

$$CLV = AOV \times \text{Purchase Frequency} \times \text{Customer Lifespan}$$

Example: If a customer spends 1,500 BDT per order, purchases 4 times per year, and remains a customer for 2 years:

$$CLV = 1500 \times 4 \times 2 = 12,000 \text{ BDT}$$

The business expects 12,000 BDT in revenue from that customer.

Return on Ad Spend (ROAS)

ROAS shows how much revenue a business earns for every unit of money spent on advertising.

$$ROAS = \frac{\text{Revenue from Ads}}{\text{Ad Spend}}$$

Example: If a business spends 10,000 BDT on advertising and earns 40,000 BDT from those ads:

$$ROAS = \frac{40,000}{10,000} = 4$$

A ROAS of 4 means every 1 BDT spent on advertising generates 4 BDT in revenue.

Bounce Rate (BR)

Bounce Rate shows the percentage of visitors who leave a website after viewing only one page without continuing to another page.

$$BR = \frac{\text{Single-Page Visits}}{\text{Total Visits}} \times 100$$

Example: If 1,000 people visit a website and 400 leave after viewing only one page:

$$BR = \frac{400}{1000} \times 100 = 40\%$$

The bounce rate is **40%**.

Cart Abandonment Rate (CAR)

Cart Abandonment Rate shows the percentage of shoppers who add products to their cart but do not complete the purchase.

$$CAR = \left(1 - \frac{\text{Completed Orders}}{\text{Shopping Carts Created}} \right) \times 100$$

Example: If 100 customers create carts but only 30 complete their orders:

$$CAR = \left(1 - \frac{30}{100} \right) \times 100 = 70\%$$

So, 70% of shopping carts were abandoned.

7. The AIDA Funnel Mode

The AIDA model (Awareness → Interest → Desire → Action) is a foundational marketing framework that describes the cognitive and emotional stages a consumer passes through from first encountering a brand to completing a purchase.

The AIDA Model is a marketing framework that explains the four stages a customer passes through from first discovering a product or brand to making a purchase.

1. Awareness

Awareness is the first stage, where customers first learn about a brand or product. The business aims to attract attention through SEO, social media ads, influencer marketing, and display ads. The main goal is to reach as many potential customers as possible and make them familiar with the brand or product. At this stage, the business focuses on creating interest and encouraging customers to learn more about what it offers.

2. Interest

In the interest stage, customers become curious and start learning more about the product by visiting websites, checking features, or comparing alternatives. Businesses use product videos, informative content, email newsletters, and retargeting ads. The main goal is to keep customers interested and help them understand how the product can meet their needs. At this stage, businesses provide useful information to encourage customers to consider the product for purchase. For example, the customer compares the smartphone with other models.

3. Desire

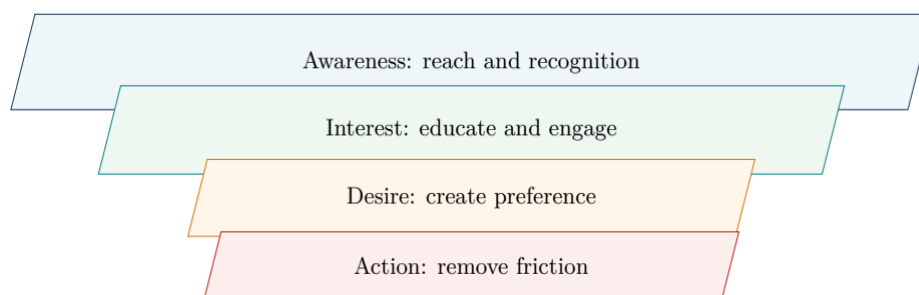
In the desire stage, customers develop a strong preference for the product but may still need convincing. Businesses use reviews, ratings, discounts, testimonials, and scarcity messages to build trust and encourage purchase. The main goal is to turn customer interest into a strong desire to own or use the product. At this stage, businesses highlight the product's benefits and value to convince customers that it is the right choice for them.

Example: A customer reads positive reviews and sees a limited-time discount on a smartphone, which increases their desire to buy it.

4. Action

In the action stage, the customer is ready to purchase. Businesses make checkout easy by providing multiple payment options, clear shipping information, trust badges, and strong call-to-action buttons. The main goal is to make the buying process simple, fast, and secure so that the customer completes the purchase. At this stage, businesses try to remove any difficulties that could cause the customer to leave without buying.

Example: A customer clicks the "Buy Now" button, selects a payment method, enters delivery information, and completes the purchase.



Strategic Importance

The AIDA model helps e-commerce businesses understand the customer buying journey and identify where customers are being lost. Different marketing strategies can then be applied at each stage to move customers toward purchase.

8. E-Commerce Fulfillment Models

E-Commerce Fulfillment refers to the process of storing inventory, processing orders, packing products, shipping them, and delivering them to customers after an online order is placed. Choosing the right fulfillment model is important because it affects cost, delivery speed, scalability, operational control, and customer satisfaction. There are four major fulfillment models:

1. In-House Warehousing / First-Party Fulfillment

In in-house warehousing or first-party fulfillment, the merchant owns or operates its own warehouses, keeps products in its own inventory, and directly manages packing and deliver orders to customers. The main advantage of this model is that the merchant has full control over inventory, product quality, packaging, branding, and shipping, which also allows faster responses when problems occur. However, it requires a high investment in warehouses, inventory, employees, equipment, and other operational resources. This model is best suited for established businesses with large and predictable order volumes or businesses dealing with fresh and perishable products. For example, Chaldal uses warehousing infrastructure to store and fulfill grocery orders.

2. Dropshipping

In dropshipping, the merchant does not keep products in its own inventory. Instead, when a customer places an order, the merchant forwards the order information to a supplier, and the supplier ships the product directly to the customer. Therefore, the basic flow is Customer → Merchant → Supplier → Customer. This model requires low initial investment because the merchant does not need to maintain a warehouse or purchase large amounts of inventory, and it is easy to add new products and scale the business. However, it usually provides lower profit margins and gives the merchant less control over product quality, inventory availability, packaging, and delivery speed. Dropshipping is best suited for new businesses, niche online stores, and businesses that want to test new products without investing heavily in inventory.

3. Third-Party Logistics (3PL)

In third-party logistics or 3PL, the merchant gives its products to an external logistics company that handles the fulfillment process. The 3PL provider stores the merchant's products in its warehouse and manages activities such as packing, shipping, delivery, and returns when customers place orders. The basic flow is Merchant → 3PL Warehouse → Customer. This model reduces the merchant's need to build and maintain its own logistics infrastructure and provides access to professional logistics services. As a result, the merchant can focus more on important business activities such as marketing and product development. A well-known example is Amazon FBA (Fulfillment by Amazon), where Amazon provides fulfillment services for sellers.

4. Fourth-Party Logistics (4PL)

In fourth-party logistics or 4PL, a business gives responsibility for managing and coordinating its overall logistics and supply chain to an external company. Instead of simply storing and delivering products, the 4PL provider manages different warehouses, 3PL providers, transportation services, delivery companies, and technology systems to ensure that the entire supply chain operates efficiently. The key difference is that 3PL performs logistics activities, while 4PL manages and coordinates the entire logistics system. The main benefit of 4PL is centralized management and optimization of a complex supply chain, making it particularly useful for large businesses that work with multiple logistics providers. For example, Alibaba's Cainiao Network uses technology and data to coordinate multiple logistics providers and improve the overall delivery network.

3PL vs 4PL

3PL performs logistics operations, such as warehousing, packing, and delivery.

4PL manages the overall logistics network, coordinating multiple logistics providers.

9. Inventory Management and EOQ

Inventory Management is the process of ordering, storing, tracking, and controlling a company's stock. Its goal is to ensure that enough inventory is available when needed while keeping inventory costs as low as possible.

EOQ (Economic Order Quantity) is the best quantity of inventory to order each time so that the total ordering cost and holding cost are minimized.

EOQ Formula

$$EOQ = Q^* = \sqrt{\frac{2DS}{H}}$$

Where D = annual demand, S = ordering cost per order, and H = annual holding cost per unit.

Total inventory cost can be calculated as:

$$TC(Q) = \frac{D}{Q}S + \frac{Q}{2}H = \text{Ordering Cost} + \text{Holding Cost}$$

The Reorder Point (ROP) tells a company when it should place a new order:

$$ROP = \text{Daily Demand} \times \text{Lead Time} + \text{Safety Stock}$$

Example

Suppose D = 1,200 units/year, S = 500 BDT/order, and H = 50 BDT/unit/year.

$$EOQ = \sqrt{\frac{2(1200)(500)}{50}} = \sqrt{24,000} \approx 155$$

So, the company should order approximately 155 units each time.

Orders per year = $1,200 \div 155 \approx 7.74$, so the company will place about 8 orders per year.

At the **EOQ (155 units)**, we can calculate the two main inventory costs:

Ordering Cost:

$$\begin{aligned}\text{Ordering Cost} &= \frac{D}{Q} \times S \\ &= \frac{1200}{155} \times 500 \approx 3,871 \text{ BDT}\end{aligned}$$

Here, $1200 \div 155 \approx 7.74$ orders per year, and each order costs 500 BDT.

Holding Cost:

$$\begin{aligned}\text{Holding Cost} &= \frac{Q}{2} \times H \\ &= \frac{155}{2} \times 50 = 3,875 \text{ BDT}\end{aligned}$$

Here, $155 \div 2 = 77.5$ units is the average inventory, and each unit costs 50 BDT per year to hold.

10. Credit/Debit Card Processing

A credit/debit card payment is processed in two main phases:

- 1. Authorization** — checks whether the payment can be approved.
- 2. Clearing & Settlement** — actually transfers the money.

Phase 1: Authorization

Authorization happens within a few seconds. At this stage, the system checks whether the transaction should be approved or declined; the final transfer of funds happens later.

Steps

1. Customer enters card details

The customer enters card information on the merchant's checkout page.

2. Payment Gateway

The payment gateway securely captures and encrypts the card information and sends it to the payment processor.

3. Payment Processor → Card Network

The processor routes the request to the appropriate card network, such as Visa or Mastercard.

4. Card Network → Issuing Bank

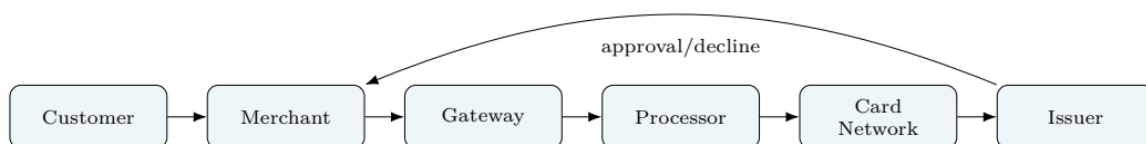
The card network sends the request to the customer's issuing bank, which issued the card.

5. Bank checks the transaction

The issuing bank checks factors such as available funds/credit, card status, account status, and possible fraud.

6. Approve or Decline

The issuing bank sends an Approved or Declined response back through the network.

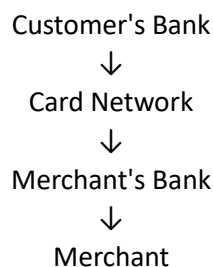


Phase 2: Clearing and Settlement

After authorization, clearing and settlement handles the actual movement of money.

Clearing: The authorized transactions are submitted for processing, and the card network determines the amounts and applicable fees.

Settlement: Funds move through the banking system to the merchant, minus applicable processing fees. The customer's transaction is also posted to their account.



Settlement commonly takes one or more business days, depending on the payment system and merchant arrangement.

11. Cyber Fraud in MFS

Cyber fraud in Mobile Financial Services (MFS) refers to fraudulent activities where criminals use digital methods or deception to steal money or sensitive information from mobile financial accounts. In Bangladesh, services such as bKash and Nagad are widely used, making MFS security an important concern. Many attacks rely on social engineering, where criminals manipulate users rather than directly hacking the system.

1. Common Types of MFS Fraud

Vishing (Voice Phishing): Fraudsters call victims pretending to be MFS customer-service representatives. They may claim that the account has a problem and ask for the user's PIN or OTP.

Smishing (SMS Phishing): Fraudsters send fake SMS messages, such as “*You have won 50,000 BDT,*” and ask the victim to provide personal information, click a malicious link, or contact a fraudulent number.

Phishing: Attackers create fake websites, emails, or messages that appear to come from trusted organizations. When users enter their credentials, the attackers steal them.

Advance Payment Fraud: Customers are asked to pay in advance for products or services that are never delivered. This can occur in fraudulent or deceptive online-commerce schemes.

2. Countermeasures

MFS providers can use multiple security measures to reduce fraud:

- **Multi-Factor Authentication (MFA):** Uses more than one authentication factor, such as a PIN plus possession of a registered device/SIM or an OTP where applicable.
- **Data Protection:** Sensitive credentials should be securely stored and protected rather than kept as readable plaintext.
- **Real-Time Fraud Monitoring:** Systems analyze unusual transactions and behavior to identify and block suspicious activities.
- **Digital KYC:** Verifies a customer's identity using official information, helping prevent anonymous or fraudulent accounts.

3. Legal and Regulatory Protection

Bangladesh has cybersecurity, financial, and consumer-protection laws and regulations that can apply to online and MFS-related fraud. Victims may report fraudulent transactions to their MFS provider and, depending on the case, relevant law-enforcement or consumer-protection authorities.

12. RFM Model

The RFM Model is a customer segmentation method used to analyze purchasing behavior and develop suitable marketing strategies. RFM stands for Recency, Frequency, and Monetary Value. Customers are usually scored from 1 to 5 on each factor.

Three Dimensions

1. **Recency (R):** Measures how recently a customer made a purchase. Customers who purchased recently are more likely to respond to new offers.
2. **Frequency (F):** Measures how often a customer buys products within a certain period. Customers who buy frequently are usually more loyal and valuable to the business.
3. **Monetary Value (M):** Measures how much money a customer has spent. It helps identify high-value and low-value customers.

RFM Segments and Strategies

Champions: Have high Recency, Frequency, and Monetary scores. They buy recently, frequently, and spend the most. **Strategy:** Give VIP rewards, early access, and referral benefits.

Loyal Customers: Purchase frequently and usually spend well. **Strategy:** Offer loyalty programs, premium products, and upselling opportunities.

Potential Loyalists: Purchased recently but have lower Frequency or Monetary scores. **Strategy:** Encourage repeat purchases through loyalty rewards and special offers.

Promising: Usually recent or first-time buyers with low Frequency. **Strategy:** Provide second-purchase discounts and engaging content.

At Risk: Were active customers but have not purchased recently. **Strategy:** Use personalized discounts and win-back campaigns.

Can't Lose Them: Previously frequent and high-spending customers who are now inactive. **Strategy:** Use premium offers and personal outreach to recover them.

Hibernating: Have low RFM scores and rarely purchase. **Strategy:** Use low-cost re-engagement campaigns.

Lost: Have the lowest RFM scores and have been inactive for a long time. **Strategy:** Invest minimally or provide a final discount.

Strategic Importance

RFM helps businesses apply targeted marketing instead of the same strategy for every customer. It improves customer retention, marketing efficiency, and ROI by focusing resources on the most valuable customer groups.

13. TPS, DSS, MIS, EIS, MRS, DPs

These are different types of Information Systems used at different levels of an organization. Some handle daily transactions, while others help managers make decisions.

1. TPS — Transaction Processing System

A Transaction Processing System (TPS) is an information system that collects, records, processes, and stores data from the daily routine transactions of an organization. It mainly supports the operational level of management, where regular business activities are performed. TPS handles a large number of repetitive transactions quickly and accurately, such as sales, payments, orders, payroll, deposits, and withdrawals. It also keeps business records updated and provides data that can be used by other information systems.

Example: In a supermarket, when a customer purchases a product, the TPS records the product name, quantity, price, and payment, generates a receipt, and automatically updates the inventory.

TPS transactions follow **ACID properties**:

Atomicity – complete fully or not at all; **Consistency** – data remains valid; **Isolation** – transactions do not interfere; **Durability** – completed transactions remain saved.

Easy meaning: TPS = Handles daily business transactions.

2. DSS — Decision Support System

A Decision Support System (DSS) is a computer-based information system that helps managers make better decisions by analyzing data, comparing different alternatives, and providing useful information or recommendations. It is mainly used for semi-structured and unstructured decisions, where the best solution is not always clear. A DSS combines data, analytical models, and software tools to help managers understand problems and evaluate possible solutions. It supports human decision-making rather than replacing the decision-maker.

Example: A sales manager uses a DSS to analyze past sales, customer demand, costs, and market trends to decide how much inventory should be ordered for the next month.

3. MIS — Management Information System

A Management Information System (MIS) is a computer-based system that collects data from systems such as TPS, processes and summarizes it, and provides useful information and reports to managers. It is mainly used by middle-level managers to monitor business performance and make regular and structured decisions. MIS helps managers compare actual results with targets, identify problems, track performance, and support planning and control.

Example: A sales manager uses MIS to check monthly sales reports, compare sales with targets, and plan future activities.

4. EIS — Executive Information System

An Executive Information System (EIS), sometimes called an Executive Support System (ESS), provides highly summarized and important information to top-level executives. It is designed for CEOs, directors, senior executives, and other strategic decision-makers. EIS usually displays information through dashboards, graphs, charts, KPIs, trends, and alerts, allowing executives to quickly understand the overall condition of the organization. Unlike MIS, EIS may combine internal information with external information, such as competitor activity, economic conditions, industry trends, and market changes.

For example, a CEO may use an EIS dashboard to see total sales, profit, market trends, and business performance.

5. MRS — Management Reporting System

A Management Reporting System (MRS) is an information system that takes business data and generates useful reports for managers. It is often considered an important part of MIS and mainly supports middle management in monitoring and controlling organizational activities.

There are three common types:

- **Scheduled Report:** Generated regularly, such as a daily sales report.
- **Exception Report:** Generated when something unusual happens, such as inventory falling below a minimum level.
- **Demand Report:** Generated when a manager specifically requests it.

6. DPS — Data Processing System

A Data Processing System (DPS) is a computer-based system that collects, processes, stores, and maintains routine organizational data. It mainly focuses on handling large amounts of data accurately and efficiently and keeping business records up to date, rather than supporting complex managerial decisions.

Example: A payroll system processes employee attendance, salary, bonuses, and deductions to generate monthly salary records.

How They Work Together

These systems can support one another within an organization:

Daily Data / Transactions



TPS / DPS



MIS and MRS



DSS



EIS



Strategic Decisions

14. The DIKW Hierarchy

The DIKW Hierarchy is an important concept in Management Information Systems (MIS). It explains how simple raw facts can gradually be transformed into useful understanding that helps people make good decisions. DIKW stands for: Data → Information → Knowledge → Wisdom

As we move upward, the content becomes more meaningful and more useful for decision-making.

1. Data — Raw Facts

Data is the lowest level of the DIKW hierarchy. It consists of raw facts, numbers, symbols, or observations that have not yet been organized or explained. By itself, data usually has little meaning because we do not know the context.

Example: 4500, 3825, 4100

These numbers are data. We do not know whether they represent sales, salaries, customers, or something else.

2. Information — Meaningful Data

Information is data that has been processed, organized, classified and placed in a meaningful context so that it becomes understandable and useful to the user. Unlike raw data, information has meaning, relevance, and purpose, and it can help people understand a situation, identify changes or patterns, reduce uncertainty, and make decisions.

Example: “Sales in Region A fell by 15% in Q3 compared with Q2.” Now the sales data has context. We know what happened, where it happened, and when it happened.

3. Knowledge — Understanding the Information

Knowledge is created when we understand information, recognize patterns, and combine it with experience, skills, or expertise. Knowledge helps us understand how something works or how different pieces of information are connected.

Example:

A doctor sees that a patient has a high temperature and an increased white blood cell count. Based on medical knowledge and experience, the doctor understands that these signs may indicate an infection. The test results are information, while understanding what those results mean is knowledge.

4. Wisdom — Making the Best Decision

Wisdom is the highest level of the DIKW hierarchy. It is the ability to apply knowledge, experience, judgment, values, and understanding to make the best decision or choose the most appropriate action in a particular situation. It goes beyond simply knowing facts or understanding patterns; it involves deciding what should be done, why it should be done, and what consequences may result.

Wisdom mainly answers Why? and What should we do?

Example:

An experienced CEO knows that rapid market growth may not continue forever. Based on knowledge of previous market cycles, the CEO decides not to expand the company too quickly during a temporary boom. The CEO is not simply understanding the situation; they are using that understanding to make a good long-term decision.



15. Information System Hierarchy

The Information System (IS) Hierarchy shows how different information systems support different levels of management in an organization. At the bottom, systems deal with detailed daily transactions. As information moves upward, it becomes more summarized and strategic for higher-level decision-making.

There are four main levels:

1. TPS — Transaction Processing System

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Example: A sales manager uses a DSS to analyze past sales, customer demand, costs, and market trends to decide how much inventory should be ordered for the next month.

3. MIS — Management Information System

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Example: A sales manager uses MIS to check monthly sales reports, compare sales with targets, and plan future activities.

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For example, a CEO may use an EIS dashboard to see total sales, profit, market trends, and business performance.

16. ERP Implementation in a Manufacturing Firm

ERP (Enterprise Resource Planning) is an integrated information system that connects different departments of an organization through a single system and shared database. In a manufacturing firm, ERP connects departments such as Manufacturing, Finance, Sales, Procurement, and Human Resources. When data changes in one department, the relevant information becomes available to other departments, reducing duplicate data and information silos.

Main ERP Modules in Manufacturing

- 1. Manufacturing:** Manages production planning, materials, schedules, factory operations, and quality control. It helps produce products on time with the required materials.
- 2. Finance & Accounting:** Manages the company's financial activities, including accounts payable and receivable, general ledger, expenses, budgeting, and financial reports.
- 3. Sales & Distribution:** Manages customer orders, pricing, inventory availability, shipping, and delivery. It connects customer demand with production and inventory.
- 4. Procurement:** Manages suppliers and purchasing activities, including purchase orders and buying raw materials needed for production.
- 5. Human Resources (HR):** Manages employee information, payroll, attendance, recruitment, training, and performance.

ERP Implementation Process

ERP implementation is not only a technical project; employees must also adapt to new processes. A firm can manage the change using Kotter's 8-Step Change Model: Create Urgency → Build a Guiding Coalition → Form a Vision → Enlist Support → Enable Action → Generate Short-Term Wins → Sustain Acceleration → Institute Change

In simple terms, management should explain why ERP is needed, create a strong implementation team, set clear goals, train employees, remove barriers, demonstrate early success, and finally make ERP part of normal business operations.

Benefits of ERP in Manufacturing

ERP provides several important benefits:

- **Integrated information:** All departments share a common database.
- **Real-time visibility:** Managers can quickly see inventory, production, sales, and financial information.
- **Better coordination:** A sales order can automatically affect inventory, production, procurement, and finance.
- **Lower cost and errors:** Reduces duplicate data entry and manual work.
- **Better decision-making:** Managers receive accurate and updated information.

Case Study: Berger Paints Bangladesh – ERP and MIS

Background

Berger Paints Bangladesh has factories, depots, and distribution networks across the country. Before using ERP, departments like Finance, Sales, Inventory, and HR used separate systems, so their information was not connected.

Problem

Because the systems were separate, financial closing took about 15 days. Inventory information was also 1–2 days old, so sales managers could not know the current stock when taking orders. This sometimes caused wrong promises to customers and customer dissatisfaction.

MIS Solution

Berger implemented SAP ERP, which connected all departments in one system. When the sales team enters an order, the system automatically updates inventory, prepares the dispatch order, updates accounts, and shows the information on the management dashboard.

Results

- Financial closing reduced from **15 days to 2 days**.
- Inventory accuracy increased from **85% to 99%**.
- Order fulfillment time reduced by **40%**.
- Managers can now see real-time information on dashboards instead of waiting for weekly reports.

Lesson

ERP-based MIS helps a company connect departments, reduce errors, save time, and get real-time information. Most importantly, it helps managers make faster and more accurate decisions.

17. NPV Calculation

Net Present Value (NPV) is used to measure whether an investment or project is financially profitable. It converts all future net benefits into their present value and then subtracts the initial investment. If $NPV > 0$, the project is financially acceptable. If $NPV < 0$, the project is generally not financially attractive at the chosen discount rate.

Formula

$$NPV = \sum_{t=1}^n \frac{B_t}{(1+r)^t} - C_0$$

Where B_t = net benefit in year t , r = discount rate, t = year, and C_0 = initial investment.

Example

Given:

Initial investment = 8,000,000 BDT

Annual net benefit = 2,500,000 BDT

Duration = 5 years

Discount rate = 10%

First, convert the discount rate:

$$r = 10\% = \frac{10}{100} = 0.10$$

Therefore:

$$1 + r = 1 + 0.10 = 1.10$$

The basic formula for each year is:

$$PV = \frac{\text{Annual net benefit}}{(1+r)^t}$$

Now calculate the present value:

$$PV = \frac{2,500,000}{(1+0.10)^1} + \frac{2,500,000}{(1+0.10)^2} + \frac{2,500,000}{(1+0.10)^3} + \frac{2,500,000}{(1+0.10)^4} + \frac{2,500,000}{(1+0.10)^5}$$

$$PV = \frac{2,500,000}{1.1} + \frac{2,500,000}{1.1^2} + \frac{2,500,000}{1.1^3} + \frac{2,500,000}{1.1^4} + \frac{2,500,000}{1.1^5}$$

$$PV = 2,272,727 + 2,066,116 + 1,878,287 + 1,707,534 + 1,552,304$$

$$PV \approx 9,476,968 \text{ BDT}$$

Finally:

$$NPV = 9,476,968 - 8,000,000$$

$$NPV \approx 1,476,968 \text{ BDT}$$

Since $NPV > 0$, the project is financially acceptable.

18. Kotter's 8-Step Change Model

Kotter's 8-Step Change Model is a framework used to manage organizational change and employee resistance when introducing new systems or technologies. In MIS, it is especially useful during major projects such as ERP implementation, because successful implementation depends not only on technology but also on employees accepting and using the new system.

1. Create a Sense of Urgency

Make employees understand why the change is needed and why it should happen now. Management should explain the problems with the current system, risks of not changing, and benefits of the new system. Example: Show that competitors are processing orders faster using modern systems.

2. Build a Guiding Coalition

Create a strong team of managers, leaders, and employees from different departments to lead and support the change. The team helps solve problems, explain the benefits, and encourage employees to accept the new system. Example: Create a team with members from IT, Finance, HR, Sales, and Operations.

3. Form a Strategic Vision

Create a clear vision and goals for what the organization wants to achieve through the change. Employees should understand the expected improvements and how the new system will benefit the organization.

4. Enlist a Volunteer Army

These supportive employees can help and motivate others who may be unsure or resistant to the new system. They clearly explain the vision to everyone in the organization and encourage employees who support the change to take part.

5. Enable Action by Removing Barriers

Identify and remove problems that prevent employees from adopting the new system. Barriers may include lack of training, outdated procedures, insufficient resources, or technical problems.

6. Generate Short-Term Wins

Create and show small successes early when introducing the new system. For example, if a new ERP module reduces costs in one department, management can share this success with employees to build trust and confidence in the new system.

7. Sustain Acceleration

Do not stop after the first success. Continue improving the system, collecting feedback, and expanding the change to other departments.

8. Institute Change

Finally, make the new system a permanent part of the organization by updating procedures, employee responsibilities, training, and performance measures.

Application in MIS: Kotter's model is particularly useful during ERP implementation. When a company replaces separate departmental systems with one integrated ERP system, employees may resist new

processes and responsibilities. Kotter's model helps management introduce the change gradually, gain employee support, provide training, demonstrate benefits, and make the new system part of everyday operations.

19. Balanced Scorecard (BSC)

The Balanced Scorecard (BSC) is a strategic performance management framework that measures an organization's performance from different areas instead of focusing only on financial results. The Balanced Scorecard has four main perspectives: Financial, Customer, Internal Process, and Learning & Growth.

1. Financial Perspective

The Financial Perspective measures whether the organization's strategies and activities are producing good financial results. It focuses on profitability, revenue growth, cost control, efficient use of resources, and return on investment. Strong financial performance shows that the organization is creating value for its owners or shareholders.

Example: A company sets a goal to increase annual profit by 10% while reducing operating costs.

2. Customer Perspective

The Customer Perspective measures how well the organization is meeting the needs and expectations of its customers. It focuses on customer satisfaction, loyalty, retention, service quality, market share, and attracting new customers. Good customer performance is important because satisfied customers are more likely to continue buying and recommend the business to others.

Example: A company aims to increase customer satisfaction to 90% by improving product quality and customer service.

3. Internal Process Perspective

The Internal Process Perspective measures how efficiently and effectively the organization's internal business processes operate. It identifies the processes that must perform well to satisfy customers and achieve financial objectives. It may focus on production efficiency, product quality, delivery speed, innovation, waste reduction, and operational improvement.

Example: A manufacturing company aims to reduce product defects and production time to improve quality and deliver products faster.

4. Learning & Growth Perspective

The Learning & Growth Perspective measures the organization's ability to develop employees, improve knowledge and skills, use technology, and support innovation. It provides the foundation for long-term improvement because skilled employees, effective information systems, and a strong organizational culture help improve internal processes and overall performance.

Example: A company provides regular employee training and introduces new technologies to improve employee skills, productivity, and innovation.

20.1. GDSS — Group Decision Support System

A Group Decision Support System (GDSS) is an information system that helps a group of people discuss problems, generate ideas, evaluate alternatives, and make decisions together. It makes group decision-making faster, more organized, and less influenced by social pressure.

Features of GDSS

- **Electronic Brainstorming:** All participants can enter ideas simultaneously, allowing more ideas to be collected quickly.
- **Anonymous Input:** Members can give opinions without revealing their identities, encouraging honest participation.

- **Idea Organization:** Idea organization helps the group arrange, categorize, and prioritize a large number of ideas. Similar ideas can be grouped together so that the team can focus on the most useful options.
- **Voting and Ranking:** Members can vote for or rank alternatives, and the system calculates the results automatically.
- **Parallel Processing:** Many participants can contribute at the same time instead of waiting for their turn.

Techniques Used in GDSS

- **1. Nominal Group Technique (NGT):** A structured technique where group members generate ideas individually, then share and discuss them with the group. Finally, members vote or rank the ideas to select the best solution. This ensures everyone gets an equal chance to contribute.
- **2. Delphi Method:** A technique where a group of experts gives opinions anonymously through several rounds of questionnaires. After each round, participants receive a summary of the group's responses and can revise their opinions. The process continues until a general agreement or consensus is reached.

20.2. Features of EIS — Executive Information System

An Executive Information System (EIS), also known as an Executive Support System (ESS), is designed for top-level executives such as CEOs and directors. It provides summarized and important information to help them monitor organizational performance and make long-term strategic decisions.

Features of EIS

- **Drill-Down:** Allows executives to move from summarized information to more detailed information, such as total sales → region → product.
- **Exception Highlighting:** Automatically highlights unusual results or important problems that require management attention.
- **Trend Analysis:** Trend Analysis shows how important business measures change over time. It helps executives identify whether sales, profit, market share, or other indicators are increasing or decreasing.
- **Graphical Display:** Presents information through dashboards, charts, graphs, and other visual formats for quick understanding.
- **Customizable:** Dashboards can be customized according to each executive's responsibilities and information needs.
- **External Data Integration:** Combines internal data with market trends, competitor information, economic conditions, and other external data.
- **Ad-hoc Analysis:** Allows executives to explore data and answer unexpected questions without relying only on predefined reports.
- **Mobile Access:** Enables executives to monitor important business information through smartphones and tablets from anywhere.

21. EIS Performance Monitoring

EIS (Executive Information System) Performance Monitoring is a system that helps top-level managers such as CEOs, directors, and board members quickly understand how well the entire organization is performing. Instead of showing thousands of detailed transactions, EIS filters and summarizes important information into KPIs, charts, alerts, and dashboards so executives can identify problems and make strategic decisions quickly.

How EIS Monitoring Works: The basic idea is: **Raw Data → Filter & Aggregate → KPIs → Dashboard → Detect Problems → Drill Down → Decision**

For example, a bank may process millions of transactions. The CEO does not need to see every transaction. An EIS might instead show:

ATM Uptime: 98.7% 

Loan Default Ratio: 7.2% 

Deposits Growth: +4.5% 

This immediately tells the executive where attention is needed.

Core Mechanisms

1. Filter and Aggregate: EIS removes unnecessary operational details and presents only important Key Performance Indicators (KPIs) such as revenue, profit, market share, loan ratios, or customer growth.

2. Internal + External Data: It combines information from inside the organization, such as TPS and MIS data, with external information such as competitor activity, economic conditions, market prices, and industry trends.

3. Near-Real-Time Monitoring: Modern EIS dashboards can update frequently, helping executives identify major changes or problems without waiting for periodic reports.

Important Features

1. Exception Highlighting

- Automatically identifies performance that differs from targets.
- Uses indicators such as red, amber, and green to show performance status.
- Helps executives quickly identify areas requiring attention.

2. Drill-Down Capability

- Starts with highly summarized information.
- Allows executives to move from general information to detailed information.
- Example: Total Revenue → Division → Region → Product → Transaction.
- Helps identify the root cause of a problem.

3. Trend Analysis

- Shows how important KPIs change over time.
- Helps identify increasing or decreasing performance patterns.
- Supports forecasting and long-term strategic decisions.
- Example: Market share changing from **27% → 26% → 25% → 24%** indicates a declining trend.

4. Graphical Dashboards

- Presents information using charts, gauges, heat maps, and graphs.
- Makes complex information easier to understand.
- Allows executives to understand organizational performance at a glance.

Example: Dutch-Bangla Bank

At Dutch-Bangla Bank, executives can use EIS-style dashboards to monitor daily deposits, loan ratios, ATM uptime, and other important KPIs. If the Non-Performing Asset (NPA) ratio shows an amber warning, management can drill down from the overall bank level to division → branch → account to identify where the problem is occurring and take appropriate action.

Strategic Frameworks

Balanced Scorecard: EIS can support performance monitoring across four areas: Financial, Customer, Internal Process, and Learning & Growth.

Critical Success Factors (CSFs): EIS can focus monitoring on the limited number of areas that are essential for an organization to achieve its strategic objectives.

22. Enterprise Resource Planning (ERP)

ERP (Enterprise Resource Planning) is an integrated information system that connects different departments of an organization through a single system and shared database. In a manufacturing firm, ERP connects departments such as Manufacturing, Finance, Sales, Procurement, and Human Resources. When data changes in one department, the relevant information becomes available to other departments, reducing duplicate data and information silos.

1. Core ERP Modules and Shared Data

A typical ERP system consists of several interconnected modules that automate specific functional areas while communicating with each other through a central database:

- 1. Manufacturing:** Manages production planning, materials, schedules, factory operations, and quality control. It helps produce products on time with the required materials.
- 2. Finance & Accounting:** Manages the company's financial activities, including accounts payable and receivable, general ledger, expenses, budgeting, and financial reports.
- 3. Sales & Distribution:** Manages customer orders, pricing, inventory availability, shipping, and delivery. It connects customer demand with production and inventory.
- 4. Procurement:** Manages suppliers and purchasing activities, including purchase orders and buying raw materials needed for production.
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ERP Implementation Process

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In simple terms, management should explain why ERP is needed, create a strong implementation team, set clear goals, train employees, remove barriers, demonstrate early success, and finally make ERP part of normal business operations.

2. Strategic Value

ERP systems allow a firm to achieve the benefits of both centralization (consistent data and economies of scale) and decentralization (empowered, real-time decision-making at the local level). By creating a "single source of truth," ERPs transform how an organization operates, significantly increasing decision speed and accuracy.

3. Implementation and Market Context

- **Market Leaders:** SAP and Oracle are the dominant global ERP vendors. Large-scale implementations can cost between USD 5 million to 50 million and take 1 to 3 years, with a typical ROI achieved within 3 to 5 years.
- **Bangladesh Context:** Prominent users of ERP systems in Bangladesh include Walton, Square Group, PRAN-RFL Group, and Berger Paints.

- **Managing Change:** Because implementation is often met with human resistance, firms frequently use Kotter's 8-Step Change Model to manage the transition by creating urgency, building a guiding coalition, and generating short-term wins.

23. The CIA Triad

The CIA Triad is a fundamental information security framework used to protect information and computer systems. CIA stands for Confidentiality, Integrity, and Availability. In e-commerce, Authenticity and Non-repudiation are also important for securing online transactions.

1. Confidentiality

Confidentiality means protecting information from unauthorized access, viewing, or disclosure. Only people who have proper permission should be able to access sensitive information. Its main purpose is to protect the privacy and secrecy of data. Organizations maintain confidentiality using usernames and passwords, authentication, encryption, access control, and user permissions.

Example: In a hospital, a patient's medical records should only be accessible to authorized doctors, nurses, and staff. An ordinary employee or outsider should not be able to view them.

If confidentiality is violated, sensitive information such as passwords, customer records, financial information, or business secrets may be exposed.

2. Integrity

Integrity means keeping information accurate, complete, consistent, and trustworthy throughout its life cycle. Data should not be changed, deleted, or damaged without proper authorization. If an authorized person changes the data, the change should be correct and properly recorded. Integrity can be protected through access controls, data validation, checksums, digital signatures, backups, and audit logs.

Example: Suppose a customer's bank balance is \$5,000. An unauthorized person should not be able to change the balance to \$50,000. Only valid transactions performed through the banking system should change the balance.

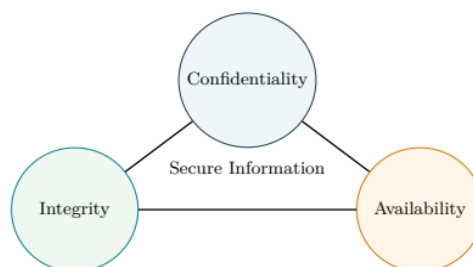
If integrity is violated, the organization may make decisions using incorrect or manipulated information.

3. Availability

Availability means ensuring that information, systems, and services are accessible to authorized users whenever they need them. A system is not useful if it is secure but unavailable when users need to access it. Availability can be maintained through backup systems, redundant servers, reliable networks, regular maintenance, disaster recovery plans, and backup power supplies. Organizations also need protection against cyberattacks that attempt to make systems unavailable.

Example: A bank needs its ATMs and online banking system available 24/7. If the main server fails, a backup server can continue providing services so customers can access their accounts.

If availability is violated, users may be unable to access important systems or information, causing business interruption, financial loss, and customer dissatisfaction.



24. Expected Value (EV)

Expected Value (EV) is the probability-weighted average payoff of a decision. It is used when there are several possible outcomes and the probability of each outcome is known. EV helps managers compare alternatives and choose the option with the highest expected return.

Formula

$$EV = \sum_{i=1}^n P_i V_i$$

Where P_i = probability of outcome i and V_i = value or payoff of outcome i .

The probabilities must always add up to 1 (or 100%):

$$\sum P_i = 1$$

Example

For **Product A**:

High demand = 0.50×6.0 million

Medium demand = 0.30×2.5 million

Low demand = $0.20 \times (-1.0)$ million

$$EV(A) = (0.50 \times 6.0) + (0.30 \times 2.5) + (0.20 \times -1.0)$$

$$EV(A) = 3.0 + 0.75 - 0.20 = \boxed{3.55 \text{ million BDT}}$$

For **Product B**:

High demand = 0.30×8.0 million

Medium demand = 0.40×3.0 million

Low demand = $0.30 \times (-2.0)$ million

$$EV(B) = (0.30 \times 8.0) + (0.40 \times 3.0) + (0.30 \times -2.0)$$

$$EV(B) = 2.4 + 1.2 - 0.6 = \boxed{3.00 \text{ million BDT}}$$

Since $3.55 > 3.00$, the company should choose Product A because it has the higher expected value.

For a decision tree, calculate EV at each chance point and work from right to left, choosing the alternative with the highest EV.

Exam Tip: Always check that the probabilities sum to 1. EV is useful for comparing expected financial outcomes, but it does not fully consider risk tolerance, worst-case outcomes, ethics, or strategic goals.

25. Simon's Model

Simon's Model of Decision Making, explains decision-making as a structured process consisting of four phases. In Management Information Systems (MIS), different information systems support managers throughout these phases to make better decisions.

1. Intelligence Phase – “What is the problem?”: In this phase, managers identify and understand a problem or opportunity by collecting and analyzing information from the organization and its environment. MIS reports, EIS dashboards, and monitoring systems can help identify unusual situations or changes. For example, a report showing an 18% decrease in regional sales may alert management that a problem needs attention.

2. Design Phase – “What are the options?”: In this phase, managers develop, analyze, and compare possible solutions to the identified problem. Decision Support Systems (DSS), what-if analysis, simulations, and financial models can help predict the possible results of different alternatives. For

example, management may consider reducing prices, increasing advertising, or offering discounts to improve sales.

3. Choice Phase – “Which option is best?”: In this phase, managers evaluate the available alternatives and select the most suitable solution. They consider factors such as cost, benefits, risks, and expected results. DSS optimization models, decision matrices, and sensitivity analysis can support this selection. For example, management may choose increasing digital advertising because it is expected to improve sales at a reasonable cost.

4. Implementation Phase – “How do we execute and monitor?”: In this phase, the selected solution is put into action and its performance is monitored to determine whether it achieves the expected results. TPS can support operational activities, while MIS reports and dashboards can monitor performance and KPIs. For example, after launching the advertising campaign, management monitors sales and revenue to check whether the decision was successful.

Key Concept: Bounded Rationality

A key part of Simon's theory is Bounded Rationality. It means that people cannot always make perfectly rational decisions because they have limited time, information, and ability to process large amounts of information.

Satisficing vs. Optimizing: Optimizing means checking all possible options and choosing the absolute best one. However, managers usually cannot examine every option because of limited time and information. Instead, they often choose the first option that is good enough to meet their needs. This is called satisficing.

MIS Implication: Information Systems help managers make better decisions by processing large amounts of data, performing complex calculations, and comparing many alternatives quickly. In this way, MIS reduces some human limitations and helps managers make more informed decisions.

26. Anthony's Triangle

Anthony's Triangle is a management framework that divides decision-making activities in an organization into three levels: Strategic Planning, Management Control, and Operational Control. It shows how decisions differ from top management to lower-level management and how information systems support each level.

1. Strategic Planning — Top Level: This is the highest level of the triangle and focuses on long-term goals and major decisions for the organization. These decisions are usually made by senior executives and are often unstructured because they involve uncertainty and future conditions. Information is usually summarized and may come from both inside and outside the organization.

Example: Deciding to enter a new international market or build a new factory.

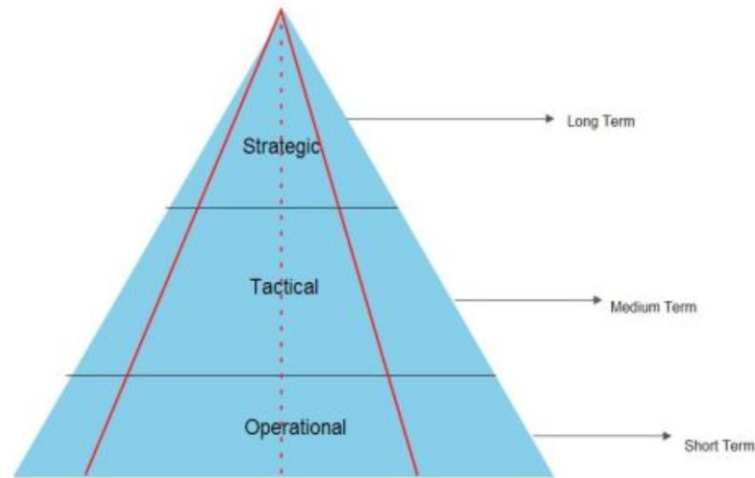
2. Management Control — Middle Level: This level focuses on putting the organization's plans into action and using resources properly. Middle managers make decisions about budgets, employee performance, use of resources, and department activities. These decisions are usually semi-structured and are supported by MIS and DSS.

Example: Creating a yearly sales budget or deciding how to distribute resources among departments.

3. Operational Control — Lower Level: This is the lowest level and focuses on daily routine activities and short-term decisions. Operational managers and supervisors ensure that specific tasks are completed correctly and efficiently. These decisions are usually structured and repetitive and are often supported by TPS.

Example: Processing customer orders, managing daily inventory, scheduling employees, or recording sales transactions.

Exam Tip: Anthony's Triangle is often used to justify the design of different information systems. For example, a CEO (Strategic) needs a graphical dashboard to see market trends, whereas a clerk (Operational) needs a fast, reliable system to process individual customer orders.



27. Decision Types by Structure

In Management Information Systems (MIS), decisions are classified into three types based on how clearly the decision-making process can be defined: Structured, Semi-Structured, and Unstructured Decisions.

1. Structured Decisions

Structured decisions are routine, repetitive, and well-defined decisions. There are clear rules, procedures, or formulas for making these decisions, so the same process can be followed whenever a similar situation occurs. Because the steps are known in advance, these decisions can often be automated by information systems. They are most common at the operational level and are often supported by Transaction Processing Systems (TPS).

Example: A company automatically places a new order when inventory falls below 100 units. The rule is already defined, so little human judgment is required.

2. Semi-Structured Decisions

Semi-structured decisions are decisions where some parts follow clear rules, while other parts require human judgment and experience. Information systems can provide data, analysis, forecasts, and possible alternatives, but the manager usually makes the final decision. These decisions are common at the middle-management level and are often supported by Management Information Systems (MIS) and Decision Support Systems (DSS).

Example: A marketing manager decides how to divide the annual advertising budget among social media, television, and other channels. The system can provide costs, past performance, and forecasts, but the manager uses judgment to make the final choice.

3. Unstructured Decisions

Unstructured decisions are new, complex, and non-routine decisions for which there are no fixed rules or standard procedures. They usually involve uncertainty and require managers to use their knowledge, experience, judgment, creativity, and intuition. These decisions are mainly made by top-level managers and may be supported by DSS and Executive Information/Support Systems (EIS/ESS).

Example: A company is considering entering a new international market. Management must consider competition, economic conditions, customer demand, risks, investment requirements, and future opportunities. There is no fixed formula that can make the final decision.

28. Network Effects

Network effects (also called network externalities) occur when a product or service becomes more valuable as more people use it. The value of the network to each individual user increases with every new user who joins the platform.

1. Metcalfe's Law

Metcalfe's Law states that the value of a network grows with the square of the number of users (n^2). As more users join, more connections become possible, making the network more valuable.

Example: With 10 users, value $\approx 10^2 = 100$. With 20 users, value $\approx 20^2 = 400$. So, doubling the users can make the network about four times more valuable.

2. Types of Network Effects

There are three main types of network effects: **Direct, Indirect (Two-Sided), and Data Network Effects.**

1. Direct Network Effects: This happens when a service becomes more useful as more people use the same service. For example, a telephone network is not useful with only one user. As more people join, users can communicate with more people, making the network more valuable. Examples: WhatsApp, Facebook, and telephone networks.

2. Indirect (Two-Sided) Network Effects: This happens when a platform connects two different groups, and more users on one side make the platform more valuable to the other side. For example, on Daraz, more sellers provide more products, which attracts more buyers. More buyers then attract more sellers.

3. Data Network Effects: This happens when more users create more data, which helps a company improve its service. Better services then attract more users and generate even more data. For example, Google can use user data to improve search results, while Amazon can improve product recommendations.

3. Strategic Implications

Network effects strongly influence competition in digital markets. As a platform gains more users, it often becomes more valuable and can gain an advantage over competitors.

1. Winner-Takes-All (or Most): As more users join a platform, it becomes more useful and attracts even more users. This can lead to one or a few companies becoming dominant in the market. For example, Google has become a dominant search engine.

2. Lock-in and Switching Costs: Network effects can make it difficult for users to leave a platform because they may lose connections, data, or useful services. For example, leaving Facebook may mean losing easy access to your existing social network.

3. First-Mover Advantage: A company that enters a market early can build a large user base before competitors, giving it a strong advantage. For example, bKash entered Bangladesh's mobile financial services market early and built a large network of users and agents.

4. Multi-homing: This means users use multiple competing platforms at the same time, which reduces dependence on one platform. For example, a driver may work for both Uber and Pathao, making it harder for one platform to completely control the market.

29. Winner Takes All

In digital markets, Winner-Takes-All (or Winner-Takes-Most) means that one or a few companies become very dominant and capture most of the market. This often happens because of network effects, data advantages, and customer lock-in.

1. Positive Feedback Loop

Network effects are a major reason for Winner-Takes-All markets. As more users join a platform, the platform becomes more valuable, which attracts even more users. Eventually, the leading platform can become difficult for smaller competitors to catch.

2. Key Drivers of Dominance

Data Moats: Large platforms collect huge amounts of user and transaction data. They can use this data to improve recommendations, AI systems, and services, giving them an advantage over new competitors.

Lock-in and Switching Costs: Platforms can make it difficult for users to switch because they may lose stored data, social connections, preferences, or connected services. For example, users may stay with a platform because their contacts, files, purchase history, or other information are already there.

First-Mover Advantage: A company that enters the market early can build a large user base, strong brand, and network before competitors, making it harder for new companies to compete.

3. Real-World Examples

Companies such as Google in search, Amazon in e-commerce, bKash in mobile financial services, and Uber in ride-sharing illustrate how digital platforms can build strong market positions through network effects and other competitive advantages.

In simple terms: More users → More value → More users → Stronger market position.

30. Metcalfe's Law

Metcalfe's Law is a principle of network economics that states that the value of a network is proportional to the square of the number of connected users (n). In simple terms, as more users join a network, the network becomes much more valuable because users can connect with more people.

1. Formula and Logic

The formula is:

$$V \propto n^2$$

Where V = network value and n = number of users.

As the number of users increases, the number of possible connections increases rapidly. For example, 2 users can make 1 connection, 5 users can make 10 connections, and 100 users can make 4,950 connections.

2. Calculation Example

Suppose a network has **10 users**, and the number of users doubles to **20**.

Original value: $10^2 = 100$

New value: $20^2 = 400$

Therefore, when the number of users doubles, the network value becomes 4 times greater.

CT-01

Question 1

Define MIS. Distinguish MIS from other elements of a system.

Answer:

MIS (Management Information System) is an organized system that collects, processes, stores, and provides useful information to managers for planning, decision-making, monitoring, and control.

A Data Processing System (DPS) mainly processes and records routine day-to-day transactions, while MIS uses processed data to produce meaningful information for management.

Basis	MIS	Data Processing System (DPS)
Purpose	Supports management and decision-making	Processes routine transactions
Output	Summarized reports and useful information	Detailed transaction records
Users	Mainly managers	Mainly operational staff
Focus	Planning, monitoring, and control	Daily business operations
Example	Monthly sales performance report	Recording daily sales transactions

Question 2

Using Anthony's Triangle, explain how TPS, MIS, DSS, and EIS support different management levels.

Answer:

Anthony's Triangle divides management into three levels:

- **Operational Level – TPS:** Handles routine daily transactions such as sales, payroll, and inventory.
- **Management Level – MIS & DSS:** MIS provides regular reports, while DSS helps managers analyze problems and make semi-structured decisions.
- **Strategic Level – EIS:** Provides summarized information to top executives for long-term planning and strategic decisions.

Thus, information becomes **more summarized and strategic** as it moves upward.

Question 3

A bank plans to migrate from a legacy MIS to a cloud-based system.

(a) Identify THREE potential risks during/after implementation.

1. **Data security risk:** Sensitive banking information may be exposed to cyberattacks.
2. **Data migration risk:** Data may be lost, corrupted, or incorrectly transferred.
3. **System downtime:** Migration may interrupt banking services and operations.

(b) Identify THREE Critical Success Factors that will determine implementation success.

1. **Strong top-management support** and proper planning.
2. **Employee training and change management** to help users adopt the new system.
3. **Strong security, testing, and backup procedures** to ensure safe and reliable migration.

CT-02

Question 1

An e-commerce company's online sales are increasing, but profits are decreasing due to high marketing and delivery costs. Analyze the situation using Value Chain, CAC, CLV, and Outbound Logistics. Suggest suitable improvements.

Answer:

Although sales are increasing, high marketing and delivery costs are reducing profit.

- **Value Chain:** Analyze marketing, storage, and delivery activities to identify and reduce unnecessary costs.
- **CAC (Customer Acquisition Cost):** Cost of gaining a new customer. Reduce CAC through targeted ads, SEO, and referrals.
- **CLV (Customer Lifetime Value):** Total value a customer generates over time. Increase CLV through loyalty programs, repeat purchases, and good customer service.
- **Outbound Logistics:** Includes packing, shipping, and delivery. Reduce costs through route optimization, bulk shipping, and better courier agreements.

Question 2

Explain the concepts of Disintermediation and Re-intermediation in E-Commerce. How can both occur simultaneously in the same industry? Use suitable real-world examples.

Answer:

Disintermediation means removing traditional middlemen and selling directly to customers.

Example: A manufacturer sells products through its own website instead of using retailers.

Re-intermediation means introducing new digital intermediaries between sellers and customers.

Example: Sellers use Daraz or other online marketplaces to reach customers.

Both can occur together. For example, hotels can accept direct bookings through their websites (**disintermediation**) while also using Booking.com as a digital intermediary (**re-intermediation**).

Conclusion: E-commerce can remove traditional middlemen while creating new digital intermediaries.

Question 3

Daraz experiences a huge traffic spike during its “11.11 Campaign.” Analyze how CDN, Load Balancer, Horizontal Scaling, and Database Read Replicas help Daraz maintain performance and availability. Also explain what problems may occur if they are not used.

Answer:

During the 11.11 campaign, millions of users may access Daraz at the same time, creating heavy server load.

- **CDN:** Delivers static content such as images from nearby servers. Without it, pages may load slowly.
- **Load Balancer:** Distributes user requests among multiple servers. Without it, a server may become overloaded or crash.
- **Horizontal Scaling:** Adds more servers to handle increased traffic. Without it, the system may not have enough capacity.
- **Database Read Replicas:** They use copies of the main database to handle user requests, reducing the load on the main database and improving website speed.

Conclusion: These technologies keep Daraz **fast, scalable, and available** during high traffic.

Question 4

Why are network effects important for digital platforms? Calculate using Metcalfe’s Law: if a platform has 15 users and later grows to 30 users, by what factor does the network value increase?

Answer:

A **network effect** means a platform becomes more valuable as more users join. More users create more possible connections, making the platform more useful and attractive.

According to **Metcalfe’s Law**:

$$V \propto n^2$$

For 15 users:

$$V_1 = 15^2 = 225$$

For 30 users:

$$V_2 = 30^2 = 900$$

Therefore,

$$\text{Increase Factor} = \frac{900}{225} = \boxed{4}$$

Conclusion: When users increase from **15 to 30**, the network value increases by **4 times**.